

Allicin as a Complementary Medicine of Triple/Quadruple Therapy for *Helicobacter pylori*: A Systemic review and Meta-analysis of Randomized Controlled Trials (protocol)

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Abstract

Backgrounds: Allicin (allyl 2-propene-1-thiosulfinate or diallyl thiosulfinate), a garlic compound, had been proved to be one of the active ingredients of anti-Hp effects. In recent years, a series of clinical trials further explored the anti-Hp effect of allicin as a complementary therapy were published with variable results.

Aim: This study aims to systemic review the allicin as a Complementary medicine of Triple/Quadruple Therapy for Hp infection and further assess such efficacy based on previous randomized controlled trials. We also aim to assess the quality of current evidences of allicin as complementary therapy of Hp infection

Methods: Electronic databases, including Medline, Embase, Web of Science, Cochrane Central Register of Controlled Trial, Wanfang Database, and VIP Database, will be searched using the keywords of “allicin”, “*Helicobacter pylori*”, “randomized clinical trials” and their synonyms. A funnel plot may be adopted to evaluate the publication bias. The meta-analysis may be performed using fixed effect model for low heterogeneity and randomized effect model for high heterogeneity. Publication bias will be evaluated by Egger’s test, and funnel plot as well if enough ($n > 10$) studies are included. What’s more, Trial Sequential Analysis (TSA) will be performed to evaluate whether the required information size has been met in order to make a determination of treatment benefit, harm or futility. The results of the meta-analysis will also be evaluated using the GRADE method and whether degradation is considered in terms of risk of bias, inconsistency, indirectness, accuracy, and publication bias, divided into “high quality”, “moderate quality”, “low quality”, and “very low quality”.

Conclusions: Our review will provide an evidence of moderate quality that allicin as a complementary medicine for *Helicobacter pylori*.

Keywords: allicin; *Helicobacter pylori*; randomized clinical trials

1. Introduction

Helicobacter pylori (Hp) is a gram-positive and microaerophilic bacterium that colonizes the gastric mucosa.[1] Globally, >50% of people are infected, and the overall prevalence is higher in developing countries[2]. Hp is one of the major factors inducing pathological changes such as gastritis and digestive ulcer, as well as atrophy, intestinal metaplasia, intraepithelial neoplasia, and mucosa-associated lymphoid tissue lymphoma (MALT) [1,2].

Proton pump inhibitors (PPIs) in combination with antibiotics have been used for eradicating Hp. PPI triple therapy (PTT), which forms of PPI and two antibiotics such as Amoxicillin, Clarithromycin and etc., was recommended as first line therapy and adopted

worldwide[2]. However, in recent decades, reports from different areas showed that the eradication rate of PTT declined to about 70% or lower[2]. Antibiotic resistance, especially for Clarithromycin and metronidazole, was accepted to be the main cause of the decline [3,4]. To further increase the eradication rate of Hp, bismuth containing quadruple therapy (BCQT) is presently recommended to be main empirical therapy in regions with high clarithromycin and metronidazole dual resistance (>15%) [1]. However, the eradication rates using BCQT ranged from 70% to 94%, still not well accepted according to previous studies. [1,2,5] Garlic (*Allium sativum* L.), one of the most widely grown vegetable crops in Asia, is known as a medical plant worldwide because of containing of 33 kinds of sulfur compounds such as allicin, alliin, ajoene, diallyl trisulfide (DATS) and others. [6] Previous studies in vitro showed that garlic might inhibit growth and colonization of bacteria as well as Hp [7]. However, a series of clinical trials that garlic oil or fresh oral garlic was medicated around 2000s failed to hold the evidence of garlic on Hp infection. [8,9] Allicin (allyl 2-propene-1-thiosulfinate or diallyl thiosulfinate), a garlic compound, had been proved to be one of the active ingredients of anti-Hp effects. [10] With development of pharmaceutical technology, commercial products of allicin tablets came to be available. Researchers improved allicin therapy as complementary to PTT and BCQT to treat Hp infection. From then on, a series of clinical trials further explored the anti-Hp efficacy of allicin published with variable results. [10,11] Based on these status, we performed this meta-analysis aiming to: (1) To systemic review the allicin as a Complementary of Triple/Quadruple Therapy for Hp; (2) To further evaluate the anti-Hp efficacy of allicin based on previous studies; (3) To assess the quality of current evidences of allicin as complementary therapy of Hp infection.

2. Methods

Information sources and search strategy

We plan to perform a systemic literature search in the databases of MEDLINE, EMBASE, Web of Science, Cochrane Database, China National Knowledge Infrastructure Database (CNKI), Chinese VIP Information (VIP), Chinese Medical Databases (CMB), and Wan-Fang Database from the starting date of database establishment till June 1st, 2019. Academic journals, dissertations, and conference proceedings will be included irrespective of gray literature status. The search terms included “*Helicobacter pylori*”, “allicin”, “randomized clinical trials” and their synonyms. The search strategies for electronic databases are listed in Appendix A. The reference lists of studies will be searched for potentially relevant titles.

Inclusion and exclusion criteria

The following criteria will be used for literature selection and the studies that meet the criteria will be eligible for inclusion.

1. The subjects enrolled were adults with Hp infection with/without Hp-related diseases such as gastritis, ulcer, etc. The diagnosis of Hp infection was in accordance with guidelines from the “Fifth Chinese National Consensus Report on the management of *Helicobacter pylori* infection” [1].
2. Subjects in the treatment group underwent interventions using allicin plus PTT or BCQT while in control group of PTT/BCQT alone.
3. The PTT/BCQT regimens in both groups should be identified.
4. The main outcome was eradication rate. The secondary outcome could be rate of side effects, relief of digestive symptoms (remission rate), and ulcer healing (healing rate and total effectiveness rate).
5. The study design consisted of randomized controlled clinical trials.

Studies that met the following criteria will be excluded.

1. Duplicate articles or evaluating the same sample.
2. Articles published as abstracts or protocols no matter their publication types
3. Studies recruited children and in vitro studies.

Study selection and data collection

All retrieved trials will be screened by two reviewers independently. Titles and abstracts will be screened in all relevant articles. Full text screening may be performed for further assessment according to the inclusion and exclusion criteria. Disagreements shall be resolved by discussion or by consulting another specialist.

Two reviewers independently extract data from included RCTs. Authors, publication year, sample size, intervention in groups and outcomes will be included. A form was built for data extraction.

Risk of bias assessment

We plan to evaluate the risk of bias of included article using Cochrane Handbook tools for Systematic Reviews of Interventions. [12] The methodological quality will be assessed from the following aspects of random sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessments, incomplete outcome data, selective reporting, and other bias. The risks were categorized as low, high, or unclear.

Statistical analyses

The meta-analysis will be conducted using the Comprehensive Meta-analysis software (version 2.2.064; Biostat Inc, Englewood, USA). A P value of <0.05 is considered significant. Statistical heterogeneity will be assessed by I^2 statistic. If I^2 values of $\geq 50\%$ with significance set at $P < 0.10$, significant heterogeneity is identified. Trials showing clinical heterogeneity will be combined according to the random effect model. Conversely, the fixed effects analysis will be used to achieve the analysis. The potential publication bias will be analyzed by Egger's test. A funnel plot may be drawn as well when including studies are sufficient ($n \geq 10$).

Trial Sequential Analysis and Evidence Quality Evaluation

Trial Sequential Analysis (TSA) is planned to evaluate whether the required information size has been met in order to make a determination of treatment benefit, harm or futility using Trial Sequential Analysis V.0.9.5.10 Beta (Copenhagen Trial Unit, Copenhagen, Denmark). [13] What's more, We also plan to evaluate the results of the meta-analysis using the GRADE as outlined by the GRADE Working Group. [14]

3. Discussion

Allicin was identified as the main active antimicrobial agent provided by garlic since 1944 [15] and was thought to be one of the active agents of anti-Hp effect [16]. Although some studies are thought to be supporting evidence, the exact mechanism is unclear up till now. [16] A series of clinical trials was performed in infectious subjects especially Hp infection.

Two previous meta-analysis [17,18] on this issue hold identical views, however, with common insufficient that limited their quality as clinical evidences. Both analyses [17,18] included one study of PPI-Clarithromycin-bismuth versus that plus allicin [19] and Hu et al. [17] included another study of Ranitidine plus allicin versus allicin alone [20]. Both therapeutic regimens were not widely accepted with suggestions of latest Guidelines and might cause unneglected bias. What's more, Hu et al. [17] performed an adequate searching with some detections missed according their searching strategy. In addition, the participants of both analyses were

patients with Hp related duodenal ulcers, which narrowed its application as evidence. Considering these statuses, we aim to perform a systemic review with higher quality. What's more, this review will be the first systematic review of Hp treating by allicin to using the GRADE system as quality assessment. We will further assess the quality of current evidences.

4. Conclusion

Our review will provide an evidence whether allicin help treating Hp as a complementary medicine. What's more, we will also support an assessment of current evidences in this issue.

Reference

1. Liu WZ, Xie Y, Hong LU, et al. Fifth Chinese national consensus report on the management of Helicobacter pylori infection. *J Dig Dis*. 2013;14:211-221.
2. Hunt RH, Xiao SD, Megraud F, et al. Helicobacter pylori in developing countries. World Gastroenterology Organisation Global Guideline. *J Dig Dis*. 2011;12:319-326.
3. Ranjbar R, Chehelgerdi M. Genotyping and antibiotic resistance properties of Helicobacter pylori strains isolated from human and animal gastric biopsies. *Infect Drug Resist*. 2018;11:2545-2554.
4. Park JY, Dunbar KB, Mitui M, et al. Helicobacter pylori Clarithromycin Resistance and Treatment Failure Are Common in the USA. *Dig Dis Sci*. 2016;61:2373-2380.
5. Lee SW, Kim HJ, Kim JG. Treatment of Helicobacter pylori Infection in Korea: A Systematic Review and Meta-analysis. *J Korean Med Sci*. 2015;30:1001-1009.
6. Jeong JH, Jeong HR, Jo YN, et al. Ameliorating effects of aged garlic extracts against Abeta-induced neurotoxicity and cognitive impairment. *BMC Complement Altern Med*. 2013;13:268.
7. O'Gara EA, Maslin DJ, Nevill AM, et al. The effect of simulated gastric environments on the anti-Helicobacter activity of garlic oil. *J Appl Microbiol*. 2008;104:1324-1331.
8. Aydin A, Ersöz G, Tekesin O, Akçiçek E, Tuncyurek M, Batur Y. Does garlic oil have a role in the treatment of Helicobacter pylori infection? *Turk Gastroenteroloji Dergisi*. 1997; 8:181-184.
9. Graham DY, Anderson SY, Lang T. Garlic or jalapeño peppers for treatment of Helicobacter pylori infection. *Am J Gastroenterol*. 1999;94:1200-1202.
10. Zhan YH, Peng NN. Allicin-Amoxicillin-Esomeprazole-Furazolidone therapy for Hp related duodenal ulcer. *Shand Med J*. 2013;53:63-64.
11. Bai HG, Jiang LZ, Wang AG. Allicin containing quadruple therapy for Hp related duodenal ulcer. *Chin Prec Med*. 2008;3:47-48.
12. Higgins JP, Altman DG, Gotzsche PC, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ*. 2011;343:d5928.
13. Wetterslev J, Jakobsen JC, Gluud C. Trial Sequential Analysis in systematic reviews with meta-analysis. *BMC Med Res Methodol*. 2017;17:39.
14. Jensen EA, Foglia EE, Schmidt B. Evidence-Based Pharmacologic Therapies for Prevention of Bronchopulmonary Dysplasia: Application of the Grading of Recommendations Assessment, Development, and Evaluation Methodology. *Clin Perinatol*. 2015;42:755-779.
15. Cavallito CJ, Bailey JH. Allicin, the antibacterial principle of Allium sativum. I. Isolation, physical properties and antibacterial action. *J Am Chem Soc*. 1944;66:1950-1954.
16. Cañizares P, Gracia I, Gómez LA, et al. Allyl-thiosulfonates, the bacteriostatic compounds of garlic against Helicobacter pylori. *Biotechnol Prog*. 2004;20:397-401.

17. Hu X, Ma JJ, Dong WG. Meta-analysis for the effects of allicin on the treatment of duodenal peptic ulcer with *Helicobacter pylori* positive. *Hainan Med J.* 2017;28:2898-2902.
18. Zhang MX, Yang J, Xu C. Allicin for *Helicobacter pylori* infection combined with duodenal ulcer: a meta analysis, *Zhej Clin Med J.* 2014;16:1135-1136.
19. Zhu YG. The efficacy of allicin combined with Rabeprazole to treat Hp infection. *China Med.* 2008;3:94-95.
20. Xue L. The efficacy of allicin combined with ranitidine to treat Hp infection. *J Mod Med Heal.* 2004;20:524.